



Kenya certificate of secondary education (K.C.S.E)

231/1
BIOLOGY
PAPER 1.
THEORY
TIME:

MARKING SCHEME

1. Interfering with:
 - Chemical mediation of synaptic junction.
 - Electrical changes of ions across the synapse.

(2mks)
2. - Endothermic animals remain active at all times despite changes in environmental tempts.
They colonize/ inhabit all habitats.

(2mks)
3. a) Transport of respiratory gases i.e. O₂ & CO₂.

b)

RBC	WBC
No nucleus	Nucleated
Disc shaped	Amoeboid.

(2mks)
4. a) Haptonasty (1mk)

b) Plants obtain nitrogen from plants when the insects are digested. (1mk)

c) Motor sensory, relay neurons (any of the three) (1mk)
5. a) K (1mk)

b) The pupil is small to allow in less light (1mk)

c) Myopia / hypermetropia / astigmatism. (1mk)
6. Magnification $= \frac{\text{Imagesize}}{\text{Actualsize}}$

$$= 50,000 = \frac{1 \times 1000}{x}$$

$$x = \frac{1000}{50,000} = \frac{1}{50} = 0.02mm$$
7. i) Ribonucleic acid (1mk)

ii) Presence of uracil as one of nitrogenous bases. (1mk)

iii) C-T-G-A-T-C-T-G-C

(1mk)
8. i) Holds the objective lenses in place enabling change from one objective lens to the other. (1mk)

ii) Concentrate light on the object on the stage. (1mk)



- iii) Regulate amount of light passing through the condenser. (1mk)
9. a) Herbivorous rej. Herbivore (1mk)
b) Absence of incisors and canines on the upper jaw. (2mks)
10. a) Golgi apparatus/ body. (1mk)
- b) i) Packaging of carbohydrates and glycoprotein.
ii) Secretion of synthesized proteins and carbohydrates.
iii) Secrete enzymes; produce lysosomes. (3mks)
11. RQ = $\frac{\text{Volume of carbon(iv) oxide}}{\text{Volume of oxygen consumed}}$
- = $\frac{9.3}{9.1} = 1.021$
- = 1.0
- b) Food is carbohydrates. (3mks)
12. Acquired characteristics / trait cannot be inherited. (2mks)
13. i) There is little accumulation of toxic wastes
ii) The waste products are formed slowly and get re-utilized by the plant.
iii) The waste products diffuse out through stomata e.g. CO₂, O₂ (3mks)
14. a) Ciliated epithelial cells (1mk)
b) Movement of trachea.
Movement of egg/ova in oviduct (1mk)
- c) Trachea
Oviduct. (1mk)
15. a) Capture recapture method. (1mk)
b) $P = \frac{Fm \times sc}{MR}$
= $\frac{435 \times 620}{75}$
= 3596 weaverbirds. (2mks)
- c) The released animals may not mix freely .
Some organism may move in and out of the study area. (1mk)
16. Hip joint (pelvic girdle and femur joint) shoulder joint (Pectoral girdle and humerus). (1mk)
17. Xylem vessels tracheids, Sclerenchyma, Collenchyma tissues



18. a) X- biceps
Y- Triceps. (2mks)
- b) X- Biceps relaxes while Y(triceps) contract. (2mks)
19. Nocturnal animals.
i) Have a high concentration of rods in the retina that are sensitive to low intensity.
ii) Retinal convergence of rods enables any little stimulation of several rods to combine and generate a nerve impulse. (2mks)
20. **Leaf A Habitat Aquatic.**
Reasons.
-Stomata only on upper surface.
-Stomata are exposed on the upper surface to hasten rate of transpiration.
-The leaf has a large surface of 30cm^2 to provide a large surface of transportation.
- Leaf B.**
Habitat. Terrestrial / normal dry land.
Reasons. Stomata are mainly on the lower surface. The upper surface is exposed to direct light and high temperature. Hence fewer stomata to reduce water loss.
-Lower/ small surface to reduce surface exposed to transpiration. (4mks)
21. Heterozygous tall X Homozygous dwarf.
Parental phenotypes : Heterozygous tall. x Homozygous Dwarf.
Parent genotype
Ganetes.
-
- Genotypes
Ratio 1Tt :1tt
(3mks)
22. Hydrogen atoms/ ions combine with carbon (iv) oxide. They form glucose using energy provided by adenosine triphosphate. (3mks)
23. i) Thigmotropism/ Haptotropism (1mk)
ii) Contact with support causes migration of auxins to the outside